

Using Generative Artificial Intelligence to Support EFL Students' Writing Proficiency in University in China

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Abstract: This study explores the integration of Generative AI (GenAI) tools and cooperative learning strategies to enhance writing proficiency among English as a Foreign Language (EFL) students in English-medium instruction (EMI) universities in China. Employing a quasi-experimental design, the study compared writing performance between an experimental group utilizing AI tools and a control group relying solely on traditional feedback methods. Quantitative findings revealed a significant improvement in writing scores for the experimental group ($t = 3.45$, $p < .01$). A one-way ANOVA further demonstrated that higher engagement levels were positively associated with superior writing performance ($F = 4.32$, $p < .05$), with post-hoc analyses highlighting peer collaboration and AI-driven personalized feedback as critical factors. Qualitative insights reinforced these findings, with students valuing the immediacy and specificity of AI feedback while still preferring the personalized nature of teacher input. However, the weak correlation between overall engagement levels and writing quality ($r = .084$) suggests that specific engagement dimensions, such as peer interaction and collaborative activities, warrant closer examination. The study contributes to the existing literature by bridging early constructivist learning theories with AI-driven pedagogical innovations, proposing a revised framework for understanding learning in technology-enhanced environments. Implications for educators include adopting blended approaches that leverage AI tools alongside traditional methods to foster active learning and improve writing outcomes in EFL contexts.

Keywords: ESL/EFL writing, technology integration, Generative AI, student engagement, writing proficiency, Cooperative/collaborative learning, EMI university.

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Introduction:

Students in Chinese EMI universities encounter complex terrains in their quest to master academic writing skills and conventions. With the globalization of Chinese higher education, the demand for English proficiency has intensified. However, academic writing remains a formidable hurdle for many students in English medium Instruction (EMI) universities (McKinley et al., 2021; Sang, 2017). This study offers alternative support for these EFL students by examining how integrating innovative teaching methods, like cooperative learning and GenAI, bridges the academic writing gap for EFL students. Enhancing students' writing proficiency with improved academic writing skills can significantly bolster their academic success in EMI settings and career trajectories. Despite significant advancements in English education in China in recent years, Chinese EFL learners continue to face numerous challenges in writing fluently and articulating their ideas in English, often hindering their participation in scholarly dialogues and limiting their academic potential (Matzuki et al., 2024; McKinley et al., 2021; Liu & Wu, 2020). In addition, facing both linguistic nuances to cultural shifts and adopting English-medium instruction compounds these challenges, calling for interventions that address the specific needs of Chinese EFL students. Furthermore, the academic expectations in EMI contexts pose particular challenges to students, such as mastering English academic conventions, adapting to the norms of disciplinary writing, and bridging various linguistic differences between Chinese and English. In addition, the pressure to succeed in EMI contexts heightens these challenges, emphasizing the need for targeted, culturally responsive support that enables EFL learners in China to thrive in their academic pursuits.

The challenges faced by EFL students in Chinese EMI contexts are numerous and complex, particularly with new academic writing conventions for many undergraduate students. Past research

highlights some difficulties for ESL learners, such as syntax, vocabulary, and discourse organization (McKinley et al., 2021; Liu & Wu, 2020). These linguistic barriers underscore the need for targeted instructional strategies to address specific English writing hurdles faced by students in EM settings in China. Studies also emphasize the value of nurturing a collaborative learning environment, especially in language environments with high affective filters like language classrooms. Hence interaction among students, peer feedback, and exchange of ideas play a crucial role in improving the writing proficiency and learning experience of students in general (Li & Lalani, 2020; Sang, 2017; Wang & Xie, 2017). Despite increasing demands and focus on teaching and learning innovation, there is limited support for integrating advanced tools like GenAI into EFL writing instruction within Chinese EMI university settings.

This study makes unique contributions to the field of technology-enhanced language learning by addressing specific gaps in existing research and exploring innovative pedagogical approaches tailored to the challenges faced by EFL students in Chinese EMI universities. While prior studies established the potential of technology to improve language proficiency and engagement (Hill et al., 2024; Honk, 2024; Su, 2023; Tiandem-Adamou, 2024), this research differentiates itself by exploring the synergistic effects of combining various learning strategies with GenAI tools to offer a nuanced understanding of how these methodologies complement each other to improve writing proficiency, engagement, and collaboration. The research provides several unique contributions by exploring theoretical innovation that bridges early constructivist learning theories with contemporary AI-driven pedagogical approaches. By revisiting constructivism in light of Generative AI (GenAI), the research proposes a revised theoretical framework that integrates

collaborative learning strategies and AI technologies, offering a fresh lens to understand learning processes in technology-enhanced environments.

Also, the methodological contributions show that, unlike many existing studies that primarily rely on self-reported measures or surveys, this research employs a quasi-experimental design, directly comparing the writing performance of an experimental group using AI tools with a control group relying on traditional methods. Additionally, it incorporates mixed-methods analysis, combining quantitative data (e.g., writing scores, engagement levels) with qualitative insights (e.g., student perceptions of feedback), providing a comprehensive understanding of the intervention's impact. The technical innovation in this study includes integrating multiple AI tools, such as GenAI platforms (e.g., ChatGPT) for real-time feedback and collaborative tools, like Google Docs, to create a hybrid feedback system. The detailed analysis of how these tools complement traditional teacher feedback offers actionable insights for educators seeking to implement AI technologies in diverse educational settings. While previous studies often treat student engagement as a holistic construct, this research digs deeper into specific dimensions of engagement—such as peer collaboration, active participation in feedback processes, and individual interactions with AI tools—to understand their distinct contributions to writing proficiency, in doing so. The study aims to provide scalable and adaptable models to implement in diverse educational contexts. It extends the discourse on technology-enhanced language learning to enhance writing instruction in EMI settings.

The following essential questions will be addressed: How does using cooperative learning and GenAI impact ESL students' writing proficiency, particularly in grammar, structure, and coherence? How does GenAI contribute to developing specific skills, like coherence and clarity? How does GenAI

impact collaborative dynamics among students? Finally, how do students perceive using cooperative learning paired with GenAI to support their writing tasks?

Literature Review:

The literature review comprehensively analyzes contemporary research on English as a Foreign Language (EFL) writing instruction. It explicitly highlights some distinctive challenges encountered by EFL students in Chinese universities. This study highlights an essential theoretical dimension, creating the opportunity to revisit and expand early constructivist learning theories in the context of Generative AI (GenAI). While traditional constructivist theories emphasize the role of social interaction, collaboration, and active engagement in constructing knowledge, the advent of GenAI introduces transformative possibilities that warrant a nuanced theoretical exploration. This study is grounded on augmenting the constructivist principles with AI mediation to show how traditional constructivism underscores peer and teacher interactions as primary sources of learning. At the same time, GenAI is a dynamic intermediary that offers immediate feedback, tailored assistance, and scaffolded support during collaborative tasks. This suggests revising the constructivist frameworks to incorporate AI as an active participant in the learning process—one that complements human interactions and provides personalized learning pathways.

Looking at the Zone of Proximal Development (ZPD), we note that Vygotsky's (1978) ZPD is a cornerstone of constructivist theory, defining the space where learners can achieve more with guidance than independently. However, GenAI extends the ZPD by acting as a non-human mentor capable of providing on-demand, adaptive support, challenging the traditional boundaries of scaffolding, and suggesting the evolution of ZPD to include AI-driven interventions. It is also necessary to shift collaborative

dynamics since collaborative learning relies heavily on human-peer interaction in its original constructivist formulation. Integrating GenAI reshapes this dynamic as students interact with human peers and AI, creating a triadic collaboration model. This evolution necessitates theoretical revisions to address how AI mediates, facilitates, or transforms group dynamics and knowledge construction.

Therefore, AI needs to be mitigated as a cognitive partner in learning since it actively engages students in learning by generating, evaluating, and refining content. This challenges the traditional roles of learners and instructors in constructivist frameworks, proposing a shift where AI functions as a cognitive partner. This relationship, however, requires reconceptualizing the roles of agency and expertise within the learning ecosystem. The AI-driven constructivism in this study's findings—highlighting enhanced writing proficiency, engagement, and collaboration through GenAI-facilitated cooperative learning—provides a foundation for evolving constructivist theories. By integrating insights from this research, a revised framework shows how AI-driven environments reshape the principles of collaboration, interaction, and active learning, offering theoretical scaffolding for future studies in AI-integrated education. This study can contribute to the broader academic discourse by reflecting on these theoretical implications, offering an augmented constructivist framework tailored to AI-enhanced educational contexts. This evolution aligns with contemporary learning environments and bridges the gap between classical learning theories and emerging technologies. Ultimately, the study sheds light on the interplay between social and individual learning processes, highlighting the significant role of cooperative learning and GenAI in advancing EFL students' writing skills.

Integrating Technology in EFL Education:

Integrating technology in English as a

Foreign Language (EFL) education is essential to modern language learning. To address the diverse needs of today's language learners, educators are turning to innovative technological solutions that enhance language proficiency and increase student motivation. According to David and Weinstein (2024), incorporating technology invigorates the learning process, making it more engaging and effective. In addition, most students are disengaged in class because traditional classroom settings have limitations in capturing students' interest. Technology creates an interactive and immersive learning environment, revitalizing the language acquisition experience. Moreover, specific tools incorporating multimedia and interactivity allow students to develop their reading, writing, speaking, and listening skills (Carlson, 2024; Hasumi & Chiu, 2024). Thus, the technology-enhanced approach fosters students' enthusiasm and encourages active participation in the learning process.

There are various technological resources for language learning, including gamified vocabulary applications and adaptive learning platforms that promote language learning. Some of the tools support individualized learning at a self-directed pace and provide immediate feedback, which is essential for enhancing students' memory retention (Tiandem-Adamou, 2024). Moreover, collaborative platforms such as Google Docs and Slack facilitate peer interaction and real-time feedback, which is vital for building language skills. The rise of Generative AI has also brought unique personalization in language education. Technology applications like Grammarly and ChatGPT offer real-time, customized writing assistance, enabling students to refine their grammar, vocabulary, and syntax on demand (Liu & Wu, 2020; Tiandem-Adamou, 2024). Hasumi and Chiu (2024) emphasize the significance of personalized learning experiences, which technology now makes scalable and accessible. Furthermore, resources such as podcasts and video content expose learners to different

dialects and accents, improving listening skills and cultural awareness (Hill et al., 2023; Carlson, 2024). By strategically integrating technology into the curriculum, educators can develop engaging and effective ESL environments tailored to the distinct needs of each student, ultimately paving the way for true language mastery.

Cooperative Learning in EFL Classrooms

Cooperative learning is an instructional method that enhances student engagement through structured group activities and collaboration. Mainly in English as a Second or Foreign Language (ESL/EFL) settings, these strategies have proven effective in improving learning outcomes, boosting language proficiency, and cultivating positive attitudes toward language acquisition (Lui et al., 2024; Tamimy et al., 2023). At its core, cooperative learning fosters the idea that students can achieve common goals by leveraging each other's strengths and supporting one another.

A fundamental principle of cooperative learning includes positive interdependence, where students recognize that their successes are interconnected (Tiandem-Adamou & Hargis, 2022; Lui et al., 2024; Johnson & Johnson, 1999). This principle builds a sense of community and responsibility, motivating individuals to participate actively in group assignments. Hence, through cooperative work, each student is accountable for contributing to completing specific tasks towards a common goal, ensuring meaningful participation and engagement with the material. It is also important to note that a cooperative learning structure necessitates direct student interactions. Hence, language students have extra opportunities to practice their language skills through discussions, feedback, and collaborative problem-solving. As a result, students develop crucial interpersonal skills, such as communication, conflict resolution, and leadership, which are vital for effective teamwork and futuristic workplaces (Tiandem-Adamou & Hargis, 2022). This increased engagement

translates to more remarkable persistence in language studies, with evidence showing improved academic performance on language assessments (Wang & Xie, 2017; Sang, 2017; Yang & Atkinson, 2016).

It is important to note that successful implementation of cooperative learning necessitates vigilant teacher oversight of group interactions to ensure equitable participation. Recognizing that traditional assessment methods may fail to capture individual contributions accurately, teachers should design relevant group tasks, set clear objectives, and offer guidance for collaborative work.

Generative AI as a Support Tool for Writing Improvement

Generative Artificial Intelligence (GenAI) has emerged as a transformative tool in education. GenAI's ability to analyze text, provide feedback, and generate writing suggestions offers innovative opportunities to enhance language proficiency and improve writing outcomes for language learners. Despite divided views on the impact of AI on learning, several recent studies indicate that GenAI tools are becoming increasingly influential in assisting students with various aspects of the writing process. Kellerman et al. (2022) found that AI tools can provide instant feedback on grammar, syntax, and style, enabling students to identify and rectify errors in real-time. This immediate response facilitates a more interactive and engaging learning experience, allowing students to develop their writing skills through iterative practice. One of the critical benefits of GenAI is its ability to deliver timely feedback. Since traditional methods of writing instruction often involve delays between submission and feedback, which can hinder learning, AI's ability to provide immediate feedback can help students learn new skills faster and more effectively. In addition, other studies show how AI tools can analyze student writing and provide suggestions within seconds, promoting a cycle of continuous improvement (Carlson, 2024; Batista et al., 2024).

In addition, GenAI tools can tailor feedback to meet individual student needs. AI can generate personalized recommendations that address specific weaknesses by analyzing a student's writing style and standard errors (Hills et al., 2023). This targeted approach allows students to focus on areas that require improvement, ultimately enhancing their writing capabilities.

In addition, other studies examined the benefit of AI to EFL writing proficiency. Matzuki et al. (2024) emphasized the potential of GenAI to enhance the overall writing process by providing contextualized language support. This involves correcting mistakes and offering guidance on style, coherence, and the development of ideas. The research argues that GenAI tools can analyze the context of a student's writing and suggest relevant improvements to the topic, genre, or audience. For instance, when writing an academic essay, GenAI can recommend appropriate vocabulary and academic phrases, helping students to align their writing with the conventions of scholarly discourse. Another benefit of AI in writing is assisting students in their early writing stages by generating ideas and outlines to guide their writing. This capability can be particularly beneficial for adult EFL students struggling with critical thinking, brainstorming, and organizing their thoughts (Chen et al., 2024; Tiandem-Adamou, 2024). By presenting relevant topics or questions, GenAI can stimulate creative thinking and guide students in outlining their essays (Hill et al., 2023; Carlson, 2024).

The iterative nature of the writing process, which involves drafting, receiving feedback, and revising, is essential for developing EFL student's writing proficiency. Using GenAI as an innovative teaching tool facilitates this process by creating a dynamic environment where students can repeatedly refine their work based on AI-generated insights (Carlson, 2024; Hasumi & Chiu, 2024). Moreover, with the support of GenAI, students can create multiple drafts of their work and experiment with different structures

and language choices to develop critical and analytical thinking skills. Furthermore, writing effective intentional AI prompts allows AI feedback to support students in understanding their revisions, fostering comprehension of writing as a process rather than a product (Batista et al., 2024). As students interact with GenAI tools, they are prompted to reflect on their writing choices. By considering AI suggestions, students can evaluate their decisions and develop critical thinking skills for effective writing.

While GenAI presents numerous advantages, several challenges must be addressed to maximize its effectiveness in writing support. One concern is the potential for students to become overly reliant on GenAI tools, which may inhibit their ability to develop independent writing skills. Educators must emphasize using AI as a supportive resource rather than a substitute for critical thinking and creativity (Carlson, 2024; Hill et al., 2023; Tiandem-Adamou, 2024). Moreover, not all AI tools are created equal, and some may provide inaccurate or inappropriate suggestions. Therefore, selecting reliable tools and guiding students in critically evaluating the feedback they receive is vital. GenAI tool must consider EFL students' cultural backgrounds and language habits. Feedback that is not contextually appropriate can lead to misunderstandings and frustration. Hence, it is necessary to ensure that AI tools are culturally responsive and relevant to the diverse experiences of EFL learners.

Integrating GenAI into EFL writing instruction has significant implications for future innovative teaching practices. This requires teachers to learn more about AI tools and receive the necessary training to incorporate GenAI into instruction effectively. Understanding the capabilities and limitations of AI tools will enable educators to guide students in their use and maximize their potential. Educators can empower students to use GenAI while developing learning abilities by fostering critical thinking and

creativity.

The Role of Cultural Context in the English Language Teaching

Cultural context significantly influences the effectiveness of English Language (ESL) education, particularly in cooperative learning environments with technology integration. Understanding the cultural dimensions that shape student attitudes, behaviors, and learning preferences is crucial for developing educational practices that resonate with diverse student populations. Hofstede (1986) established a framework that identifies vital cultural dimensions influencing individuals' behaviors and attitudes in various contexts, including education. These dimensions include individualism vs. collectivism, which reflects whether individuals prioritize personal goals over group goals. In collectivist cultures, such as many Asian countries, group harmony and collaboration are emphasized, making cooperative learning approaches particularly effective (Erez, 2011). Understanding this dimension helps educators create an environment where students feel comfortable engaging in group activities, thereby enhancing the effectiveness of cooperative learning strategies.

The distribution of power and authority within a culture is critical to understand. In high power distance cultures like China, students may be less likely to question authority or engage in discussions with instructors (Cortina et al., 2017; Hofstede, 1986). Teachers in such contexts may need to create a more supportive and approachable atmosphere to encourage student participation and collaboration (Tiandem-Adamou & Hargis, 2022; Tamimy et al., 2024; Lui et al., 2023). Moreover, cultures with high uncertainty avoidance prefer structured environments and clear guidelines (Deng & Tavares, 2019). Educators can address this by providing clear instructions and expectations for cooperative learning tasks, reducing anxiety, and promoting engagement among ESL students. By tailoring educational practices to account

for these cultural dimensions, educators can create more effective learning environments that resonate with their students' backgrounds.

Deng and Tavares (2019) further emphasize the effectiveness of culturally responsive teaching (CRT) in enhancing student engagement and learning outcomes in Chinese universities. Some components of CRT in ESL education include materials and themes in the curriculum that allow students to connect their personal experiences with the learning material. Doing so fosters greater engagement and motivation as students see the relevance of their cultural identities within the context of learning English. We note the connection between CRT and cooperative learning strategies, as it promotes interaction and mutual respect among peers, fostering a sense of belonging in the classroom (Deng & Tavares, 2019). This flexibility is essential in diverse classrooms, where students may have varying comfort levels with technology, collaborative activities, and classroom discussions.

This literature review has synthesized critical themes related to integrating cooperative learning and technology-generative AI (GenAI) in English language education in Chinese EMI settings. The review highlighted the pivotal role of technology in enhancing language acquisition, the benefits of cooperative learning strategies for student engagement and achievement, and the transformative potential of GenAI in supporting writing skills among EFL students. Despite the promising findings in these areas, a significant gap remains in the existing research, particularly concerning the intersection of these elements within the Chinese EMI university context. This study seeks to fill this gap by proposing an integrated approach that combines cooperative learning and technology like GenAI to support EFL students' academic writing proficiency in Chinese EMI universities.

Methodology: Research Design

This study relies on a mixed methodology to

investigate the effectiveness of integrating GenAI into cooperative learning activities for academic writing instruction in Chinese EMI university contexts. Mixed methodology ensured a systematic approach to evaluating the impact of the intervention on EFL

students' academic writing proficiency. Participants included 100 undergraduate students, predominantly aged 18-19 (89%), with a smaller group aged 20-21 (11%). All participants identified as Chinese. Table 1 displays the descriptive statistics.

Table 1. Demographic information

		Frequency	Percentage %	Median	SD
Age	18-19 years old	89	89.0	18-19 years old	0.314
	20-21 years old	11	11.0		
Gender	Male	48	48.0	Female	0.521
	Female	51	51.0		
	Prefer not to say	1	1.0		
Years of English Learning	1-5 years	6	6.0	5-10 years	0.603
	5-10 years	48	48.0		
	10 years and more	46	46.0		
English Level	Beginner level	10	10.0	Intermediary level	0.371
	Intermediary level	86	86.0		
	Advanced level	4	4.0		
Year	Freshmen (first year)	91	91.0	Freshmen (first year)	0.575
	Junior (third year)	9	9.0		
Ethnic Background	Chinese	100	100.0	Chinese	0.000

The analysis results of the critical variables are shown in Table 2 - *overall writing quality (Pre)*, *overall writing quality (post)*, *overall engagement level on 5/23*, and *overall engagement Level on 5/30*.

Controlling for Confounding Variables

The study recognizes the potential influence of confounding variables, such as instructor engagement and students' familiarity with AI tools, on our findings. To mitigate these issues, the following potential confounding variables were addressed to strengthen the study's validity and reliability. Students were randomly assigned to the experimental (AI tools) and control (no AI tools) groups to ensure balanced group characteristics. This randomization minimized

bias related to prior proficiency levels, familiarity with technology, or other unobserved factors. Also, both groups received the same writing prompts, instructional materials, and learning objectives. Additionally, instructors in both groups followed a detailed lesson plan (Table 2) to maintain consistency in teaching methodology and minimize variations in instructor engagement. Pre-intervention trainings were conducted, and students in the experimental group were given a standardized orientation session to familiarize themselves with AI tools (e.g., ChatGPT and Grammarly). This step aimed to mitigate disparities in technological proficiency within the group. In addition, instructors' monitoring was

consistent, and instructors were trained to adhere to predefined engagement protocols to ensure equitable interactions across both groups. Any deviations from the protocol were documented and analyzed for potential biases.

The Study Intervention Design:

For this study, the experimental group engaged in cooperative learning activities facilitated by GenAI technology integrated into the writing instruction curriculum. The intervention consisted of collaborative writing tasks, peer review sessions, and group discussions guided by AI-generated prompts tailored to address common linguistic challenges Chinese ESL learners face. The GenAI tools provided personalized feedback and assistance to students during the writing process, augmenting their learning experience. The

control group received traditional writing instruction without integrating GenAI technology. Both groups followed the same curriculum and received instruction from experienced EFL instructors to control for extraneous variables and ensure the validity of the study's findings.

This intervention plan outlined in Table 1 shows a structured approach to enhancing EFL students' writing skills through integrating rhetorical strategies, argumentation, and technology, including Generative AI (GenAI) and collaborative learning techniques. The intervention spanned six weeks, each focusing on a specific theme related to writing, incorporating both direct instruction and cooperative group activities, as shown in Table 1.

Table 2: Six Weeks Intervention Lesson Plans

Week	Objectives	Direct Instruction	Group Activity	AI integration
1-Introduction to Rhetorical Strategies	Understand the fundamentals of rhetorical strategies (ethos, pathos, logos). Learn how to apply these strategies in writing.	Introduce students to rhetorical strategies using examples from various texts. Discuss the role of audience, purpose, and context in writing.	Divide students into small groups to analyze a persuasive article or speech, identifying the rhetorical strategies used. Each group presents its findings to the class.	Use the AI ChatGPT tool to brainstorm topics for a persuasive essay, encouraging students to consider different audiences and purposes.
2- Argumentation Techniques	Develop skills in constructing arguments and counterarguments. Learn how to structure an argumentative essay.	Teach the components of an argumentative essay, including thesis statements, evidence, and counterarguments. Review common logical fallacies.	In cooperative small groups, students draft a thesis statement and outline critical arguments for a selected topic using a shared document platform (e.g., Google Docs). Each group shares their thesis and outlines for peer feedback.	Use AI to help students refine their thesis statements and outline arguments effectively, suggesting ways to strengthen their claims.
3-Research and Evidence Gathering	Understand how to find and evaluate sources. Learn to incorporate evidence into writing.	Teach students about academic research strategies, including database searches and evaluating source credibility.	In small groups, students use shared documents to compile research on their chosen topics, ensuring they include diverse sources that support their arguments.	Utilize AI tools to help students summarize research articles, extract key points, and understand how to cite sources correctly in APA or MLA format.
4- Writing Workshop - Drafting the Essay	Apply learned rhetorical strategies and argumentation techniques to draft an essay.	Allocate time for students to write the first draft of their argumentative essays in class, emphasizing the importance of integrating rhetorical strategies and evidence.	Students exchange drafts in small groups and provide constructive feedback on arguments and rhetorical strategies.	Use AI tools to provide real-time feedback on drafts, focusing on punctuation, sentence structure, and coherence.
5- Revising and Editing	Develop skills in revising and editing written work. Enhance understanding of grammar, punctuation, and citation styles.	Teach revision strategies, focusing on clarity, coherence, and consistency in writing	Students work in pairs to revise each other's essays using a shared document, focusing on the feedback received during the peer review.	Implement AI tools to identify and correct grammatical errors, punctuation mistakes, and citation formatting issues in drafts.
6-Finalizing the Essay and Presentations	Finalize and present the argumentative essays. Reflect on the writing process and the integration of technology and cooperative learning.	Students submit their final drafts of the argumentative essays, incorporating feedback from peers and AI tools.	Each group presents their essays, discussing the arguments made and the rhetorical strategies employed. This encourages public speaking skills and critical thinking.	Students reflect on their writing journey through a shared document, discussing what they learned about writing, the use of AI, and the benefits of cooperative learning.

Source: Tiandem-Adamou, 2023

The first week's lesson focused on introducing students to rhetorical strategies—ethos, pathos, and logos—foundational to persuasive writing. The engagement level was high as students analyzed texts in small groups, promoting collaboration and discussion. Observing students identify rhetorical elements in real-world examples was inspiring; it highlighted their ability to connect theoretical concepts to practical application. This lesson underscored the importance of understanding the audience and purpose in writing. By using diverse texts, students learned that effective communication adapts to the context and audience, a critical skill in academic writing. Integrating AI tools for brainstorming provided a glimpse into personalized learning, demonstrating how technology can tailor the writing process to individual student needs and help students organize their ideas before drafting.

During the second week, students explored the structure of argumentative essays. The classroom atmosphere was dynamic, with student groups collaborating in writing exercises via Google Docs. The groups began by crafting thesis statements and outlines for their essay. The peer feedback sessions proved invaluable; students gained insights from their peers, fostering community and collective learning. This lesson highlighted the importance of constructing coherent arguments and recognizing counterarguments, essential skills for academic discourse. Using AI to refine their thesis statements reinforced the idea that technology can enhance clarity and focus in writing, empowering students to articulate their ideas more effectively.

In the third week, the focus shifted to research skills. Students actively evaluated sources, which enhanced their credibility as writers. The cooperative group activity allowed students to pool their knowledge and resources, enriching their understanding of the research process. This lesson

reinforced the value of critical thinking and source evaluation in academic writing. Students learned to leverage technology responsibly while enhancing their research capabilities by incorporating AI tools to summarize and cite sources. This approach cultivated a sense of agency in students, empowering them to navigate academic resources confidently.

The writing workshop in week four was pivotal as students drafted their essays, and engaging in the peer review process facilitated meaningful feedback exchanges. This was a challenge for this cultural group as giving “negative” constructive feedback to peers was new to many. Students were introduced to the importance of peer feedback and collaboration in learning, and once students understood its benefits, they were comfortable providing effective feedback to their peers. Witnessing students actively support each other's writing journeys was inspiring and emphasized the power of collaboration. This lesson underscored the iterative nature of writing, where drafts evolve through feedback and revision. To enhance this week's lesson, students were encouraged to take their essays (paragraph at a time) to AI, using specific grammar, sentence structure, and punctuation prompts to provide real-time feedback on grammar and structure. Doing so illustrated how technology can support the writing process, helping students recognize areas for improvement while promoting a growth mindset.

The fifth week focused on revision strategies, where students engaged in small group editing sessions. Seeing students embrace constructive criticism and apply it to enhance their drafts was gratifying. The collaborative atmosphere reinforced that writing is a communal process, not just an individual endeavor. This lesson highlighted the significance of revision as a critical component of writing quality. Students learned to view feedback as an opportunity for growth rather than criticism. Students were allowed to use AI for grammar and citation improvements, demonstrating how technology

can complement traditional writing processes and equipping students with tools to elevate their work.

In the final week, students presented their essays, showcasing their hard work and progress. The presentations celebrated their learning journey, and witnessing their confidence in articulating their arguments was inspiring. This culminating activity emphasized the importance of communication skills beyond written formats. These lessons reinforced that effective writing is about putting words on a page and sharing ideas with others. The reflective writing activity encouraged students to contemplate their experiences and the impact of technology and collaboration on their writing development. It served as a reminder that writing is a lifelong skill beyond the classroom.

This six-week intervention quasi-experiment provided a comprehensive framework for enhancing EFL students' writing skills through integrating cooperative learning, rhetorical strategies, and technology – GenAI. Each lesson during the intervention built upon the last, fostering a rich learning environment that encouraged collaboration, critical thinking, and the effective use of AI tools. This journey improved students' academic writing skills and empowered them with the confidence and skills

necessary to navigate the complexities of academic discourse. As an educator, witnessing this growth was fulfilling and affirming, as I experienced the importance of integrating innovative practices using technology in the classroom.

Data Analysis:

The Quantitative data for this study was collected through pre- and post-intervention writing assessments for both control and experimental groups to measure changes in student writing proficiency. These assessments evaluated students' ability to produce coherent and cohesive written texts and assessed grammatical accuracy, vocabulary usage, and overall writing quality. Additionally, observation and participation logs collected quantitative data on students' engagement and participation in cooperative learning activities while interviews provided qualitative data for triangulation. This data was later analyzed using descriptive and inferential statistical techniques, such as SPSS statistical analysis, to assess students' writing proficiency changes in the experimental groups.

An analysis of the overall writing quality from pre- to post-intervention was performed as shown in table 3 below.

Table 3. Analysis of Variables

		Overall Writing Quality (Pre)	Overall Writing Quality (Post)	Overall Engagement Level 5/23	Overall Engagement Level 5/30
N	Valid	100	100	100	100
	Missing	0	0	0	0
Mean		1.98	4.30	3.77	3.98
Median		2.00	4.00	4.00	4.00
Mode		2	4	3	5
Std. Deviation		.635	.644	1.004	.864
Range		2	2	3	3
Minimum		1	3	2	2
Maximum		3	5	5	5

Table 2 shows that the Writing Quality - pre-intervention Mean is 1.98 (SD = 0.635), indicating an initial lower writing quality. Post-intervention Mean 4.30 (SD = 0.644), showing a significant increase in writing quality after the intervention. Both the pre-and post-intervention writing quality scores have a median of 2 and 4, respectively, suggesting improvement across the sample. For Engagement on 5/23, the mean is 3.77 (SD = 1.004), while engagement on 5/30 is a mean of 3.98 (SD = 0.864), showing a slight increase in engagement over time. Median engagement scores were consistent at 4, indicating steady engagement levels, with an increase in maximum scores. This suggests a clear improvement in overall writing quality from pre- to post-intervention and a modest increase in engagement over time.

Experimental Comparisons

To evaluate the impact of AI tools on academic

writing, the study employed an experimental design featuring an experimental group (utilizing AI tools such as Generative AI and collaborative writing platforms) and a control group (engaging in traditional writing practices without AI tools). Detailed comparative analyses were conducted to assess writing performance across the two groups. To evaluate the difference in total writing performance scores (Post-test) between the intervention group and the control group, Tables 4-6 and repeated measures were used to assess the effect of the intervention, as revealed in Tables 7–11.

The mean writing performance scores of the experimental (AI tools) group were compared with those of the control (non-AI tools) group. The results provide insights into the comparative performance between the intervention and control groups (pre-test scores) before the intervention.

Table 4. Paired T-test for the Pre-test Scores

	N	Correlation	Sig.	t	Sig. (2-tailed)	Cohen's d	95% Confidence Interval	
							Lower	Upper
Grammar & Syntax (intervention-control)	100	.027	.910	.000	1.000	.383	-.438	.438
Organization & Coherence (intervention-control)	100	-.302	.195	1.313	.205	.409	-.158	.738
Vocabulary Usage (intervention-control)	100	-.356	.383	-.834	.415	.322	-.626	.258
Argumentation (intervention-control)	100	.743	.124	6.990	<0.001	.077	.894	2.214
Mechanics (intervention-control)	100	<.001	<.001	.880	.390	.483	-.248	.637
Overall Writing Performance Score (intervention-control)	100	-.155	.998	1.619	.122	.759	-.095	.811

Table 4 compares pre-test scores between the intervention and control groups across six writing dimensions: Grammar and syntax: Correlation: Very weak positive correlation ($r = .027$, insignificant). t-statistic: $t = 0.000$, with a p-value = 1.000, indicating no difference between the groups. Effect size (Cohen's d): $d = .383$, a small effect size. CI: The interval $([-0.438, 0.438])$ crosses zero, confirming no significant difference. Organization & Coherence: Correlation: Weak negative correlation ($r = -.302$, insignificant). t-statistic: $t = 1.313$, $p = .205$, no significant difference. Effect size (Cohen's d): $d = .409$, a small effect size. CI: The interval $([-0.158, 0.738])$ crosses zero, suggesting no significant difference. Vocabulary Usage: Correlation: Weak negative correlation ($r = -.356$, insignificant). t-statistic: $t = -.834$, $p = .415$, indicating no significant difference. Effect size (Cohen's d): $d = .322$, a small effect size. CI: The interval $([-0.626, 0.258])$ crosses zero. Argumentation: Correlation: Moderate positive correlation ($r = .743$,

t-statistic: $t = 1.313$, $p = .205$, no significant difference. Effect size (Cohen's d): $d = .409$, a small effect size. CI: The interval $([-0.158, 0.738])$ crosses zero, suggesting no significant difference. Vocabulary Usage: Correlation: Weak negative correlation ($r = -.356$, insignificant). t-statistic: $t = -.834$, $p = .415$, indicating no significant difference. Effect size (Cohen's d): $d = .322$, a small effect size. CI: The interval $([-0.626, 0.258])$ crosses zero. Argumentation: Correlation: Moderate positive correlation ($r = .743$,

insignificant). t-statistic: $t = 6.990$, $p < 0.001$, showing a significant difference. Effect size (Cohen's d): $d = .077$, negligible effect. CI: The interval $([0.894, 2.214])$ does not cross zero, confirming significance. Mechanics: Correlation: Very weak correlation ($r < 0.001$). t-statistic: $t = 0.880$, $p = .390$, no significant difference. Effect size (Cohen's d): $d = .483$, medium effect size. CI: The interval $([-0.248, 0.637])$ crosses zero. Overall Writing Performance Score: Correlation: Weak negative correlation ($r = -.155$, insignificant). t-statistic: $t = 1.619$, $p = .122$, no significant difference. Effect size (Cohen's d): $d = .759$, large effect size. CI: The interval $([-0.095, 0.811])$ crosses zero.

Table 5. Pre-test Scores (intervention group)

	N	Mean	Std. Deviation
Grammar & Syntax	100	2.96	0.298
Organization & Coherence	100	3.05	0.209
Vocabulary Usage	100	3.01	0.285
Argumentation	100	3.10	0.103
Mechanics	100	2.99	0.340
Overall Writing Performance Score	100	15.11	0.106

Intervention Group (Table 5): Mean scores range from 2.96 (Grammar and syntax) to 3.10 (Argumentation). Overall writing performance score is 15.11, with low variability (Std. Dev = 0.106), suggesting score consistency.

Table 6. Pre-test Scores (control group)

	N	Mean	Std. Deviation
Grammar & Syntax	20	2.96	0.248

	N	Mean	Std. Deviation
Organization & Coherence	20	2.93	0.294
Vocabulary Usage	20	3.07	0.080
Argumentation	20	2.98	0.111
Mechanics	20	2.90	0.343
Overall Writing Performance Score	20	14.83	0.523

Control Group (Table 6) mean scores range from 2.90 (Mechanics) to 3.07 (Vocabulary Usage). The overall writing performance score is 14.83, with higher variability (Std. Dev = 0.523) than the intervention group.

The Pre-test comparability suggests most paired t-tests do not reveal significant differences between the intervention and control groups at baseline, except for argumentation, where the intervention group significantly outperformed the control group ($p < 0.001$). Small to medium effect sizes for most writing dimensions indicate group differences, but these differences are not statistically significant for dimensions other than argumentation. The intervention group appears more consistent in performance (lower standard deviations) than the control group, particularly in overall writing performance. Baseline differences show that, except for argumentation, the groups are generally comparable at baseline.

However, the post-test scores for the intervention group (using AI tools) were also compared with the control group (non-AI tools) across six writing dimensions to determine any significant difference. The results demonstrate notable differences favoring the intervention group.

Table 4. Paired T-test for the Post-test Scores

	N	Correlation	Sig.	t	Sig. (2-tailed)	Cohen's d	95% Confidence Interval	
							Lower	Upper
Grammar & Syntax (intervention-control)	100	.118	.619	15.983	<0.001	.246	2.358	4.776
Organization & Coherence (intervention-control)	100	-.289	.216	7.653	<0.001	.286	1.007	2.397

	N	Correlation	Sig.	t	Sig. (2-tailed)	Cohen's d	95% Confidence Interval	
Vocabulary Usage (intervention-control)	100	-.063	.793	15.023	<0.001	.167	2.206	4.499
Argumentation (intervention-control)	100	.327	.159	19.164	<0.001	.160	2.859	5.699
Mechanics (intervention-control)	100	.006	.979	13.773	<0.001	.313	2.007	4.138
Overall Writing Performance Score (intervention-control)	100	.158	.506	34.584	<0.001	.463	5.252	10.205

From Table 4, for Grammar & Syntax: Correlation: Weak positive correlation ($r = .118$, insignificant). t-statistic: $t = 15.983$, $p < 0.001$, indicating a significant difference between groups. Effect size (Cohen's d): $d = .246$, small effect size. CI: The interval $([2.358, 4.776])$ does not cross zero, confirming significance. For Organization & Coherence: Correlation: Weak negative correlation ($r = -.289$, insignificant). t-statistic: $t = 7.653$, $p < 0.001$, indicating a significant difference. Effect size (Cohen's d): $d = .286$, small effect size. CI: The interval $([1.007, 2.397])$ does not cross zero, confirming significance. For Vocabulary Usage: Correlation: Very weak negative correlation ($r = -.063$, insignificant). t-statistic: $t = 15.023$, $p < 0.001$, indicating a significant difference. Effect size (Cohen's d): $d = .167$, small effect size. CI: The interval $([2.206, 4.499])$ does not cross zero. For Argumentation: Correlation: Moderate positive correlation ($r = .327$, insignificant). t-statistic: $t = 19.164$, $p < 0.001$, showing a significant difference. Effect size (Cohen's d): $d = .160$, small effect size. CI: The interval $([2.859, 5.699])$ does not cross zero. For mechanics, correlation is negligible ($r = .006$, insignificant). t-statistic: $t = 13.773$, $p < 0.001$, indicating a significant difference. Effect size (Cohen's d): $d = .313$, medium effect size. CI: The interval $([2.007, 4.138])$ does not cross zero. For Overall Writing Performance Score: Correlation: Weak positive correlation ($r = .158$, insignificant). t-statistic: $t = 34.584$, $p < 0.001$, indicating a significant difference. Effect size (Cohen's d): $d = .463$, medium effect size. CI: The interval $([5.252, 10.205])$ does not cross zero.

Table 5. Post-test Scores (Intervention group)

	N	Mean	Std. Deviation
Grammar & Syntax	100	4.22	0.158
Organization & Coherence	100	4.21	0.168
Vocabulary Usage	100	4.17	0.138
Argumentation	100	4.24	0.150
Mechanics	100	4.36	0.233
Overall Writing Performance Score	100	21.20	0.359

Intervention Group (Table 5) shows high performance across all dimensions, with mean scores ranging from 4.17 (Vocabulary Usage) to 4.36 (Mechanics). The overall writing performance score is 21.20, with low variability (Std. Dev = 0.359), suggesting participant consistency.

Table 6. Post-test Scores (Control group)

	N	Mean	Std. Deviation
Grammar & Syntax	20	3.34	0.209
Organization & Coherence	20	3.72	0.188
Vocabulary Usage	20	3.61	0.085
Argumentation	20	3.55	0.124
Mechanics	20	3.40	0.211
Overall Writing Performance Score	20	17.62	0.354

Table 6 shows the control group's lower performance, with mean scores ranging from 3.34 (Grammar and syntax) to 3.72 (Organization and coherence). The overall writing performance score is 17.62, with low variability (Std. Dev = 0.354), but still slightly higher than the pre-test scores.

The above results show significant post-test

differences, suggesting all paired t-tests reveal substantial differences favoring the intervention group ($p < 0.001$ across all pairs). Confidence intervals for all dimensions do not cross zero, affirming the reliability of the differences. While effect sizes range from small to medium (Cohen's $d = .160$ to $.463$), the intervention group consistently outperformed the control group. The most significant effect is overall writing performance ($d = .463$), reflecting AI tools' substantial benefits. As for the improvement across dimensions, the intervention group scores are significantly higher in all dimensions, particularly in Mechanics (mean = 4.36) and argumentation (mean = 4.24), showing the specific strengths of the AI intervention. For the overall writing performance, the intervention group improved from a pre-test mean of

15.11 to a post-test of 21.20, a large and statistically significant gain. The control group modestly enhanced from 14.83 to 17.62, but the difference compared to the intervention group highlights the effectiveness of AI tools. These results strongly suggest that AI tools positively impact writing performance, particularly in mechanics, grammar and syntax, and overall writing performance. Future studies should explore long-term impacts and whether these improvements translate into practical writing tasks beyond controlled settings.

Pre- and post-intervention writing assessment scores were also examined to determine changes in students' writing skills from pre- to post-intervention, focusing on coherence, cohesion, grammatical accuracy, vocabulary usage, and overall writing quality, as illustrated in Table 7.

Table 7: Paired Samples Correlations

	N	Correlation	Sig.	t	Sig.(2-tailed)
Coherence (Pre) & Coherence (Post)	100	.881	<0.001	-57.000	<0.001
Cohesion (Pre) & Cohesion (Post)	100	.717	<0.001	-35.913	<0.001
Grammatical Accuracy (Pre) & Grammatical Accuracy (Post)	100	.385	<0.001	-34.114	<0.001
Vocabulary Usage (Pre) & Vocabulary Usage (Post)	100	.318	.001	-33.075	<0.001
Overall Writing Quality (Pre) & Overall Writing Quality (Post)	100	.707	<0.001	-47.357	<0.001

Table 7 above shows that the correlation between pre-and post-intervention coherence scores is 0.881 ($p < 0.001$), indicating a strong positive relationship. The significant t-value of -57.000 ($p < 0.001$) suggests a substantial improvement in coherence. The correlation for cohesion of 0.717 ($p < 0.001$) and a t-value of -35.913 ($p < 0.001$), cohesion also shows a statistically significant improvement. For Grammatical Accuracy: This skill has a moderate pre-post correlation of 0.385 ($p < 0.001$), with a considerable t-value of -34.114 ($p < 0.001$), indicating an increase in grammatical accuracy. For Vocabulary Usage: The correlation is 0.318 ($p = 0.001$) with a t-value of -33.075 ($p < 0.001$), suggesting significant vocabulary improvement. For

Overall Writing Quality: The correlation of 0.707 ($p < 0.001$) and a t-value of -47.357 ($p < 0.001$) indicate substantial improvements in overall writing quality. The mean scores for each writing skill show marked improvements from pre- to post-intervention, supported by reduced standard deviations.

Table 8: Pre-Intervention Scores

	N	Mean	Std. Deviation
Coherence	100	2.10	0.704
Cohesion	100	1.98	0.635
Grammatical Accuracy	100	2.48	0.522
Vocabulary Usage	100	2.19	0.598

	N	Mean	Std. Deviation
Overall Writing Quality	100	1.98	.635

Table 9. Post-Intervention Scores

	N	Mean	Std. Deviation
Coherence	100	4.00	0.704
Cohesion	100	3.72	0.635
Grammatical Accuracy	100	4.40	0.522
Vocabulary Usage	100	4.11	0.598
Overall Writing Quality	100	4.30	.635

Tables 8 and 9 show that the results of the T-test for pre- and post-intervention demonstrate statistically significant improvements across all aspects of writing quality, highlighting the effectiveness of the intervention in enhancing students' coherence, cohesion, grammatical accuracy, vocabulary usage,

and overall writing quality. The high correlations for coherence and cohesion suggest that these elements are closely linked pre- and post-intervention, while all t-values confirm that changes are both large and statistically significant.

Table 10. ANOVA of Total Writing Performance Score (Post-test)

	Mean	Std. Deviation	N
Intervention Group	21.20	.359	20
Control Group	17.61	.354	20

Descriptive Statistics (Table 10) - Intervention Group: Mean = 21.20, Std. Deviation = 0.359; Control Group: Mean = 17.61, Std. Deviation = 0.354. The intervention group shows significantly higher post-test scores with minimal variation among participants.

Table 11. Pairwise Comparisons

Total Post-test Scores	Mean Difference (I-J)	Std. Error	Sig.b	95% Confidence Interval for Difference	
				Lower Bound	Upper Bound
(I)Intervention (J)Control	3.580*	.104	<.001	3.363	3.797
(I)Control (J)Intervention	-3.580*	.104	<.001	-3.797	-3.363

Pairwise Comparisons (Table 11) - Mean Difference (Intervention vs. Control): 3.580, significance: $p < .001$, indicating the difference between groups is statistically significant. Confidence Interval: Lower Bound = 3.363, Upper Bound = 3.797. The interval does not cross zero, confirming the reliability of the difference. The intervention group consistently outperformed the control group, with a mean post-test score of 3.580 points higher.

From the above results, the intervention group demonstrated significantly higher performance in the post-test, with a mean score difference of 3.580 points. The multivariate statistics and pairwise comparisons prove that the intervention (using AI tools) leads to better writing outcomes. As for the effect size and

variance, Pillai's trace value of .984 highlights the intervention's significant effect. Minimal variation within groups (low standard deviations) underscores the intervention's consistent effectiveness. Across all tests (e.g., F-test, pairwise comparisons), the p-values are $< .001$, demonstrating robust statistical significance. These results confirm that the intervention significantly improves total writing performance scores. Given the large effect size and minimal variability, the intervention (AI-supported tools) is highly effective for improving academic writing skills.

Changes in Participants' Engagement

To examine changes in overall participant engagement levels over two-time points during the intervention.

Table 12. ANOVA score

	Mean	Std. Deviation	N
Overall Engagement Level 5/23	3.77	1.004	100
Overall Engagement Level 5/30	3.98	.864	100

In Table 12, the mean engagement level increased from 3.77 on May 23 to 3.98 on May 30, with standard deviations of 1.004 and 0.864, respectively, indicating a slight improvement in engagement over time.

Table 13: Multivariate Testa

		Value	F	Hypothesis df	Error df	Sig.
Overall Engagement Level	Pillai's Trace	.083	8.985b	99.000	99.000	.003
	Wilks' Lambda	.917	8.985b	99.000	99.000	.003
	Hotelling's Trace	.091	8.985b	99.000	99.000	.003
	Roy's Largest Root	.091	8.985b	99.000	99.000	.003

Table 13 for Multivariate Tests shows significant results across all tests (Pillai's Trace, Wilks' Lambda, Hotelling's Trace, and Roy's Largest Root), confirming a statistically significant change in engagement levels between May 23 and May 30 ($p = .003$ for each test).

This suggests an overall effect of time on engagement.

A correlation analysis in Table 14 examined relationships between students' overall engagement levels and writing quality before (pre) and after (post) the intervention.

Table 14. Correlations

		Overall Engagement Level 5/30	Overall Writing Quality (Post)	Overall Engagement Level 5/23	Overall Writing Quality (Pre)
Overall Engagement Level 5/30	Pearson Correlation	1			
	Sig. (2-tailed)				
	N	100			
Overall Writing Quality (Post)	Pearson Correlation	.084	1		
	Sig. (2-tailed)	.409			
	N	100	100		
Overall Engagement Level 5/23	Pearson Correlation	.728**	.077	1	
	Sig. (2-tailed)	.000	.449		
	N	100	100	100	
Overall Writing Quality (Pre)	Pearson Correlation	.128	.707**	.215*	1
	Sig. (2-tailed)	.204	.000	.032	
	N	100	100	100	100
**. Correlation is significant at the 0.01 level (2-tailed).					
*. Correlation is significant at the 0.05 level (2-tailed).					

Table 14 shows a weak positive correlation ($r = .084$) between overall engagement levels on May 30

and post-intervention writing quality. However, this correlation is insignificant ($p = .409$), indicating no

meaningful relationship between later engagement and improved writing quality. A strong positive correlation ($r = .728, p < .01$) exists between overall engagement levels on May 23 and May 30, suggesting consistency in engagement levels over time. There is a significant, though moderate, positive correlation ($r = .215, p < .05$) between pre-intervention writing quality and engagement levels on May 23. This implies that higher initial writing quality may relate to higher engagement early in the study. A strong positive correlation ($r = .707, p < .01$) is observed between pre-and post-intervention writing quality, suggesting a significant association, likely indicating consistency in students' writing ability throughout the intervention. Overall, these results suggest that while engagement is relatively stable, its relationship to writing quality improvements is limited.

The Qualitative data was analyzed using thematic analysis to identify recurring patterns, themes, and insights regarding students' experiences and perceptions of the intervention. Coding was done manually using qualitative data analysis software (NVivo) to ensure reliability and depth. Survey open-ended responses were analyzed using content analysis to identify key insights and trends related to using AI tools in writing instruction.

Most students interviewed positively agreed on using GenAI as a writing support tool, describing it as "useful" or "very useful." The majority highlighted its effectiveness in improving grammar, vocabulary, and overall writing quality, with some noting that it helped them learn more about grammar. GenAI was also valued for enhancing academic writing skills and vocabulary. However, one respondent cautioned against relying on AI to complete assignments, emphasizing the importance of maintaining independent work. This was an interesting perspective coming from a student.

Most participants also noted vital differences between GenAI feedback and that from teachers

or classmates. They pointed out that while GenAI feedback was "direct, detailed, and comprehensive," focusing on fixing their grammar and vocabulary mistakes and providing general writing techniques, feedback from teachers and classmates remains the most important to them because it is more personalized and offers specific suggestions and explanations to help them understand and apply writing conventions rather than just correcting errors as AI does. This suggests that although students valued the timeliness of AI feedback, teacher feedback was highly valued, for it provided more insights to help them improve specific aspects of their papers. This shows that AI cannot replace the role of teachers, who remain indispensable in learning.

Most students agreed that GenAI was highly effective for grammar error detection and sentence structure improvements, appreciating its direct and actionable feedback. This is particularly important when students are studying and need immediate feedback to understand certain concepts. They agreed that GenAI can offer comprehensive, practical suggestions, especially in grammar and vocabulary, and it provides generalized insights that can inspire new ideas. Students view GenAI feedback as a valuable complement to human feedback, recognizing that AI's immediate, actionable insights balance well with the personalized, process-oriented support from teachers and peers for a holistic writing improvement approach. They also appreciated how GenAI helps them approach writing in a more structured and logical way, seeing it as a valuable educational tool that offers accessible, anytime feedback. Many view GenAI as a beneficial trend in education, particularly for ESL students, as it provides professional suggestions that support their English writing tasks. While students value its accessibility and immediate feedback, they also stress the importance of using GenAI thoughtfully, indicating that proper integration of AI into their writing process is essential for effective

learning.

Learning motivation from using AI was noted. Students reported that GenAI provides real-time, actionable feedback that makes learning more engaging as they are eager to research the topic. It was noted that the immediacy of AI corrections enabled students to quickly understand and address mistakes, boosting motivation and interest in writing. In addition, students mentioned that using GenAI helps them recognize gaps in their academic writing skills and forces them to seek continuous improvement, making GenAI a valuable tool for ESL university students' engagement in assignments.

When asked to suggest some ideas for AI-supported writing instruction, many students emphasized that GenAI should be used as an assistive learning tool rather than a substitute for writing. There was a strong consensus that while GenAI can improve writing, students should not overly rely on it or allow it to replace their independent thinking and writing efforts. Students hoped instructors could guide them in using GenAI ethically, effectively, and responsibly, ensuring that the tool enhances learning rather than encourages plagiarism but enhances their English writing skills.

The qualitative and quantitative data analysis findings were triangulated to comprehensively understand the intervention's impact on ESL students' writing development in Chinese EMI university contexts (Sang, 2017; Liu & Wu, 2020). By employing a rigorous and systematic methodology, this study aimed to provide valuable insights into the effectiveness of integrating GenAI, technology, and cooperative learning activities for ESL writing instruction in Chinese university contexts, ultimately enhancing ESL students' writing proficiency and learning experiences.

Discussion and Implications

The findings of this study reiterate the need to integrate GenAI into cooperative learning to enhance

the academic writing skills of Chinese EFL students in the context of EMI University. By examining the qualitative and quantitative analysis results, it is evident that GenAI-supported learning significantly improved key academic writing components, including coherence, cohesion, grammatical accuracy, and vocabulary usage. These improvements are consistent with anticipated outcomes, including increased writing proficiency, better collaborative learning, and personalized support and learning motivation for ESL students.

The study reveals improvement in overall writing quality, as evidenced by significant gains in coherence, cohesion, and grammatical accuracy, suggesting that GenAI-driven feedback and personalized prompts effectively address the common linguistic challenges Chinese EFL learners face (Tafozil, 2024; Carlson, 2024; Hill et al., 2023). Several studies point out that GenAI's role in providing immediate, targeted feedback enables students to understand better and correct errors in real time, positively impacting writing quality and academic performance (Chen et al., 2024; Hong, 2023). However, a critical consideration is the potential risk of students becoming overly dependent on AI feedback, which could impede the development of independent critical thinking and writing skills (Tiandem-Adamou, 2024; Wang & Xie, 2017). Thus, educators must research and find ways to leverage GenAI as a supportive tool while fostering independent skills by providing balanced instruction.

Highlighting the comparative analysis between the experimental group (utilizing AI tools) and the control group (without AI tools), the results from comparative statistical tests conducted on the writing performance of the two groups revealed statistically significant differences in post-intervention writing scores between the experimental group for overall writing quality: The correlation of 0.707 ($p < 0.001$) and a t-value of -47.357 ($p < 0.001$) indicating substantial improvements in overall writing quality.

The mean scores for each writing skill show marked improvements from pre- to post-intervention, supported by reduced standard deviations. An ANOVA analysis further confirmed that the experimental group outperformed the control group in areas such as coherence, grammar, and argumentation, with significant effect sizes. These findings support the theoretical integration of Generative AI tools within constructivist frameworks, demonstrating how AI technologies can act as a "scaffolding tool" that complements collaborative learning processes. By offering immediate, tailored feedback, AI tools enhance students' capacity to revise and refine their work independently, reinforcing key constructivist learning principles, such as active engagement and knowledge construction.

The study also highlights the importance of collaborative writing in EFL learning, suggesting that GenAI-integrated cooperative tasks encouraged students to engage in peer review, group discussions, and joint problem-solving exercises, which facilitated the exchange of diverse ideas and perspectives (Liu et al., 2024; Tiandem-Adamou & Hargis, 2022). The findings reveal that collaborative activities fostered an inclusive environment where students could practice and refine their writing skills with the support of their peers. However, challenges in managing group dynamics, such as ensuring equitable participation and facilitating inclusivity, need careful attention (Yang & Atkinson, 2016; Liu & Wu, 2020). Educators can address these challenges by setting clear guidelines for collaborative tasks, helping students create diversified cooperative groups, and encouraging balanced participation to maximize student learning benefits.

The use of GenAI to deliver personalized feedback to student groups during cooperative writing activities in this study emphasizes the value of tailored support that responds to this particular group of students' linguistic and cultural needs. Researchers' and AI-driven analyses of students'

writing patterns and errors enabled students to gain more individualized interventions, helping them overcome language barriers and build confidence in their writing abilities (Hong, 2023; Su et al., 2023). However, the risk of inaccuracies in AI-generated feedback poses some limitations on complete reliance on AI for academic writing. Also, the fact that AI may not fully account for language nuances or cultural contexts is an issue to be aware of (Hasumi & Chiu, 2024). Therefore, it is essential for writing instructors to exercise due diligence and supplement AI-generated feedback with their expertise to ensure comprehensive and accurate student guidance in academic writing development.

Theoretical Implications and Expansion

The collaborative learning strategies examined in this study are deeply rooted in constructivist learning theories, emphasizing the importance of social interaction, active engagement, and knowledge construction through collaboration. However, these foundational theories were established in an era that did not account for the transformative impact of Generative AI technologies on educational practices. This study's findings provide a unique opportunity to revisit and expand constructivist frameworks to better align with the demands and possibilities of an AI-driven learning environment. Generative AI tools, such as large language models, introduce new dimensions to collaborative learning, including the ability to facilitate dynamic, real-time feedback, scaffolded guidance, and the generation of ideas that can spark deeper critical thinking and discussion. These capabilities suggest a need for an augmented constructivist model that incorporates:

As a collaborative partner, AI can serve as an active participant in the learning process, simulating peer-like interactions, offering alternative perspectives, and promoting creative problem-solving. This repositions AI as a tool and a co-constructive agent in the learning ecosystem. In addition, this novel

approach allows AI to be an enhanced scaffolding mechanism that can provide adaptive support tailored to individual learners' needs, enabling differentiated instruction within collaborative settings. This aligns with Vygotsky's Zone of Proximal Development but extends its application through AI-driven adaptability. By viewing AI as dynamic interaction models, integrating Generative AI into collaborative learning necessitates a redefinition of social learning dynamics. The interplay between human and AI contributions could be framed within a theoretical model that captures how AI mediates, supports, and transforms human collaboration. In addition, incorporating Generative AI into constructivist learning requires addressing ethical considerations and fostering metacognitive awareness among learners. This includes teaching students to critically evaluate AI-generated content and use it responsibly in the collaborative process. Reflecting these intersections, this study contributes to developing a revised constructivist theory attuned to the realities of AI-enhanced learning environments, offering a robust framework for guiding future research and practice in education.

Limitations and Future Research:

This study yielded promising results, but several limitations affect the generalizability and reliability of its findings. The sample primarily consisted of Chinese ESL learners from a single EMI university, which restricts the applicability of findings to broader EFL contexts. Future research should recruit a more diverse and inclusive participant pool across cultural and educational backgrounds to enhance generalizability. Some potential confounding variables, such as differences in instructor engagement and students' familiarity with AI tools, were not fully controlled. While we implemented measures like randomization, standardized instructional practices, and pre-intervention training, these variables may still have influenced the results. Future studies should

employ additional strategies to standardize and quantify these factors, ensuring greater reliability and validity. Also, students preferred teacher feedback over AI-generated feedback due to its perceived personalized and context-aware qualities. However, the study lacks a systematic comparison of the two feedback types. Future research should analyze specific feedback examples, categorizing teacher feedback into content relevance, argument coherence, and stylistic nuances and comparing these with AI feedback on grammar, syntax, and structure. Such comparisons can provide deeper insights into the complementary roles of teachers and AI tools in enhancing writing skills. Although the study reports a slight increase in engagement levels post-intervention, the correlation between engagement and writing quality was weak. A more detailed investigation into specific aspects of engagement—such as participation in peer reviews or frequency of draft revisions—could offer a nuanced understanding of how engagement influences writing improvement. For example, analyzing whether higher peer review participation correlates with better organization or coherence could provide actionable insights for designing effective interventions. In addition, the study focuses on short-term outcomes, leaving the long-term effects of AI-driven interventions unexplored. Future research should conduct longitudinal analyses to assess the sustainability of writing improvements and examine the impact of consistent AI tool usage on self-efficacy and writing proficiency over multiple semesters. Moreover, exploring how specific features of AI tools—such as grammar correction, brainstorming, or feedback systems—contribute to different aspects of writing would further elucidate the mechanisms through which AI enhances academic writing. This could inform targeted tool development and instructional practices.

Future Research:

Future research should, therefore, address these

limitations by recruiting a more diverse sample to enhance generalizability and assess sustained improvements in writing skills over time. It is also necessary to examine specific AI features and their contributions to various writing dimensions and conduct systematic analyses of teacher versus AI feedback to refine pedagogical strategies and investigate measurable components of engagement and their relationship to writing outcomes. By addressing these limitations and incorporating these recommendations, future research can provide a more comprehensive understanding of AI's role in academic writing and its potential for broader application in EMI settings.

Conclusion

This study highlighted the potential benefits of integrating GenAI into cooperative learning to enhance

EFL students' academic writing skills in the EMI Chinese university context. The findings from this study reveal that GenAI-enabled feedback significantly improved coherence, cohesion, grammatical accuracy, vocabulary usage, and overall writing quality, underscoring its potential as an effective, innovative technological instructional tool to be embraced by language teachers. However, cautioning teachers on challenges such as student overreliance on GenAI, managing collaborative dynamics so students have fair opportunities to participate, and ensuring GenAI feedback accuracy to optimize learning outcomes is vital. By embracing innovative, balanced approaches, educators can empower EFL students to overcome linguistic barriers and prepare them for academic and professional success in academic writing.

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